

Statistical Physics

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Capítulo 1

Solids

1.1. Solids with N atoms in 3D

$$T = \frac{1}{2}m \sum_{i=1}^{3N} \dot{x}_i^2 \quad ; \quad x_i : \text{positions of } N \text{ atoms at time } t$$

$$\hookrightarrow T = \frac{1}{2}m \sum_{i=1}^{3N} \dot{u}_i^2 \quad \text{taking } u_i(t) = (x_i - x_i^0)$$

$$V(\{x_i\}) = V(\{x_i^0\}) + \sum_{i=1}^{3N} \left(\frac{\partial V}{\partial x_i} \right)_{\{x_i^0\}} (x_i - x_i^0) + \frac{1}{2} \sum_{i,j=1}^{3N} \left(\frac{\partial^2 V}{\partial x_i \partial x_j} \right)_{\{x_i^0\}} (x_i - x_i^0)(x_j - x_j^0) + \dots$$

At the *equilibrium position*: $V(\{x_i^0\}) = 0$ and $\left(\frac{\partial V}{\partial x_i} \right)_{\{x_i^0\}} = 0$. For small oscillations, higher order terms are neglected.

$$H = \sum_{i=1}^{3N} \frac{1}{2} m \dot{u}_i^2 + \frac{1}{2} \sum_{i,j=1}^{3N} K_{ij} u_i u_j \quad (1.1)$$

where $K_{ij} = \left(\frac{\partial^2 V}{\partial x_i \partial x_j} \right)_{\{x_i^0\}} \rightarrow$ Microscopic physics of the solid (its interactions).

But $H \neq \sum H_i$ due to the potential. As our framework needs $H = \sum H_i$, we perform a transformation to decouple it.

1.1.1. Normal mode decomposition

$$u = \begin{pmatrix} u_1 \\ \vdots \\ u_{3N} \end{pmatrix}. \text{ Then } T = \frac{1}{2} m \dot{u}^T \dot{u} \text{ and } V = \frac{1}{2} u^T K u \text{ (Our original Hamiltonian).}$$

We search for an orthogonal transformation ($S^T S = I$) over $u = Sq$ such that $S^T K S = \Lambda$, where Λ is diagonal.

$$\begin{aligned}\dot{u}^T &= \dot{q}^T S^T \\ u &= Sq\end{aligned}$$

$$\text{then } H = \frac{1}{2}m\dot{q}^T\dot{q} + \frac{1}{2}q^T\Lambda q \rightarrow H = \frac{1}{2}\sum_{i=1}^{3N}m\dot{q}_i^2 + \frac{1}{2}\sum_{i=1}^{3N}\lambda_i q_i^2$$

Which can be expressed as $H = \sum H_i \rightarrow H$ becomes a sum of independent oscillators!!

$$H = \sum_{i=1}^{3N} \left[\frac{p_i^2}{2m} + \frac{1}{2}m\omega_i^2 q_i^2 \right] \text{ with } p_i = m\dot{q}_i \text{ and } \omega_i = \sqrt{\frac{\lambda_i}{m}} \quad (1.2)$$

λ_i is the eigenvalue of K .

A very complicated movement can be decomposed as a sum of normal modes.

Example 1: Eq. of motion for the displacements

$$H = \sum_{i=1}^{3N} \frac{p_i^2}{2m} + \frac{1}{2} \sum_{i,j=1}^{3N} K_{ij} u_i u_j \text{ with } p_i = m\dot{u}_i.$$

$$\dot{u}_i = \frac{\partial H}{\partial p_i} = \frac{p_i}{m} \quad \dot{p}_i = -\frac{\partial H}{\partial u_i} \rightarrow m\ddot{u}_i = -\sum_{j=1}^{3N} K_{ij} u_j \implies \ddot{u} = \frac{-K}{m} u$$

$$\text{Since } u = Sq: S\ddot{q} = -S\frac{\Lambda}{m}q \implies \ddot{q} = -\frac{\Lambda}{m}q \implies \ddot{q}_i + \frac{\lambda_i}{m}q_i = 0 \implies \ddot{q}_i + \omega_i^2 q_i = 0$$

Solution: $q_i(t) = a_i \cos(\omega_i t + \phi_i)$ $u_i(t) = (Sq)_i \rightarrow u_i(t) = \sum_{j=1}^{3N} S_{ij} a_j \cos(\omega_j t + \phi_j)$
 a_j and ϕ_j are fixed by initial conditions.

Example 2: 2 atoms (1D)

$$V(u_1, u_2) = \frac{1}{2}ku_1^2 + \frac{1}{2}c(u_1 - u_2)^2 + \frac{1}{2}ku_2^2 \quad u(t)? \text{ Initial conditions: } u_1(0) = u_2(0) = \alpha, \\ \dot{u}_1(0) = \dot{u}_2(0) = 0.$$